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Reproducing and stress-testing an oxygen extraction margin

An uncertainty analysis of an AI-assisted reconstruction of a neonatal oxygenation timeline

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ABSTRACT

Background. Large language models can now produce detailed physiological reconstructions of clinical events from sparse records. Such reconstructions look rigorous, but their conclusions may rest on unmeasured inputs. We examine one example: an AI-assisted reconstruction of the first 32 minutes after a newborn's birth, together with the "oxygen extraction margin" calculator that produced its numbers.

Methods. We extracted every quantitative claim and classed it as reported, inferred, assumed, or unknown. We re-implemented the calculator from its displayed equations and intermediate values, reproduced its outputs, and then measured how its central quantity, the ratio of extractable oxygen delivery to demand (M), responds to its inputs: analytically through elasticities, one input at a time over stated ranges, and jointly by Monte Carlo propagation.

Results. The calculator's values at reported saturations of 50% and 58% were reproduced exactly (M = 0.42 and 0.71). Its value at 94% (2.56) was not reproduced by the default inputs (2.04). Because M is proportional to a small difference between two oxygen contents, it amplifies input error: at a 50% reading, a 1% relative change in saturation, standard P_{50}, or critical venous PO_2 moves M by 4.4-5.6%. Across plausible single inputs, M at 50% ranged from 0.13 to 0.72. Under illustrative joint uncertainty, the 90% interval was 0.00-1.03 at 50%, 0.18-1.38 at 58%, and 1.24-3.11 at 94%. The estimated cumulative oxygen debt

depended chiefly on an assumption about an interval with no recorded saturations: about 86 mL O_2/kg if a deficit is assumed from minute 5, and 27 mL O_2/kg if it is counted only from the first reading. The supplied materials also disagree on what the first recorded saturation was.

Interpretation. The model is transparent and internally coherent. Its qualitative conclusion of an oxygen shortfall at the lowest reading holds across most of the explored input space, but its precise margins, its threshold, its cumulative debt, and any conclusion about neurological outcome do not follow from the available record. We propose eight reporting practices for computational reconstructions of clinical events.

KEYWORDS neonatal transition; oxygen delivery; pulse oximetry; fetal haemoglobin; sensitivity analysis; uncertainty quantification; reproducibility; large language models; clinical reasoning

> Methodological research. This paper does not assess the care any infant received, does not offer a view on any infant's outcome, and is not clinical guidance. Case details are de-identified and given in minutes after birth. Clinical records were not reviewed by the authors; every case detail is therefore *reported*.

1 Introduction

Clinical records rarely capture physiology continuously. Around a birth, observations arrive as a few time-stamped notes, a saturation reading when a probe is attached, and an intervention recorded after the fact. Reasoning about what happened between those points has always required inference.

Large language models change the economics of that inference. In minutes, they can turn a handful of observations into a narrative with oxygen contents, extraction ratios, deficits, and a verdict. The output is fluent, numerate, and often physiologically literate. It is also difficult to audit, because the assumptions that generate the numbers are rarely separated from the observations they are attached to.

This paper treats one such reconstruction as an object of study. Our question is not whether its conclusions are correct, which the available record cannot establish, but which of them are *determined by the data* and which are *determined by the model's inputs*. We answer it by reproducing the calculation, then deliberately stressing it.

Our contributions are:

1. a complete specification and independent re-implementation of the extraction-margin model used, with its outputs reproduced;
2. an analytic explanation of why this class of margin amplifies measurement and parameter error near the critical threshold;
3. one-at-a-time and joint uncertainty analyses at each reported reading;

4. a claim-by-claim evidential audit of the reconstruction; and
5. reporting practices for computational reconstructions of clinical events.

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2 Materials

2.1 Case narrative

The case concerns a term newborn. A family member supplied a narrative summary, a calculator, and a timeline table compiled from the clinical record. As reported, there is no continuous saturation record for the first 17 minutes. Around minute 17 the infant was noted to be dusky. A pulse oximeter was applied, the infant was placed prone, and chest physiotherapy was given. Room-air SpO_2 values of about 50% and about 58% are reported for the period from minute 17 to minute 27. Nasal cannula oxygen at 3 L/min began around minute 27, and SpO_2 was 94% at minute 32. The supplied materials also refer to arterial blood gas measurements taken hours later; these are outside the scope of this paper.

A discrepancy in the record. The supplied materials disagree about the first documented saturation. The calculator's timeline table lists 50% at minute 17, followed by 58% after the interventions. A separate summary of the same record gives the first documented SpO_2 entry as 58%, at the same time as the note of a dusky appearance. We could not resolve this without the primary record. We analyse both readings, but the discrepancy itself is a finding (Section 4.6).

2.2 The source reconstruction

The reconstruction was produced with the assistance of a large language model. It asserted that a hypoxic exposure of about 22-24 minutes ran from minute 5 to minute 28.5. It assumed that SpO_2 sat at about 50% throughout the unmonitored interval. It reported extraction margins, deficits, and a cumulative systemic oxygen debt of about 91 mL O_2/kg. It modelled cerebral oxygen delivery with an assumed doubling of cerebral blood flow, and it concluded that permanent brain injury was "almost certainly" absent if heart rate remained at or above 100 beats per minute. The reconstruction also stated that care fell short of resuscitation guidelines. We do not evaluate that statement (Section 5.4).

2.3 The calculator

The numbers in the reconstruction come from a web calculator, "Neonatal O_2Sufficiency: Physiological Calculator (v2)"[10]. It converts a pulse-oximeter reading into arterial PO_2 through a Bohr-adjusted Hill equation. It then defines a critical venous oxygen floor and expresses oxygen delivery above that floor as a multiple of an assumed demand. The calculator displays its inputs, its intermediate values, and preset scenarios with their margins. It replaced an earlier version (v1) whose output at the same 50% reading was reported as "5.67x SAFE"; v1 was not available to us and is discussed only as described by the reconstruction (Section 4.7).

3 Methods

3.1 Claim extraction and evidential classification

Each quantitative or causal statement in the reconstruction was recorded with its stated value and the inputs it depends on. Each was assigned one of five categories:

- ● **measured** : captured by an identified instrument;
- ○ **reported** : described in a narrative or record not reviewed by us;
- △ **inferred** : derived from other quantities by a model;
- ◌ **assumed** : introduced so that a model could run; and
- ▨ **unknown** : not establishable from the available evidence.

Because no primary record was reviewed, no case observation could be classed as measured.

Table 1. **Quantitative claims in the source reconstruction and their evidential status**

CLAIM	STATED VALUE	RESTS ON	STATUS
Duration of hypoxic exposure	22-24 min (minute 5 to ~28.5)	Assumed start at minute 5; assumed crossing time at 64%	△ Inferred from assumptions
SpO_2 during minutes 5-17	~50% plateau	One later reading, extended backwards	◌ Assumed
PaO_2 estimates	21.2, 24.4, 57.5 mmHg	Choice of dissociation curve; accuracy of reported readings	△ Model-dependent
Systemic extractable margin	0.37-0.42, 0.71, 2.56	Proposed metric; assumed oxygen consumption and cardiac output	△ Metric not validated
Systemic oxygen deficit	5.0-5.4 and 2.47 mL O_2/kg/min	Same inputs as the margin	△ Model-dependent
Cumulative systemic oxygen debt	~91 mL O_2/kg	Deficits integrated over assumed plateaus	△ Inferred from assumptions
Cerebral oxygen consumption	1.8 mL O_2/100 g/min	Population value, not measured in this infant	◌ Assumed
Cerebral blood flow	40, rising to 80-85 mL/100 g/min	Assumed compensatory vasodilation	◌ Assumed
Cerebral margin	1.02 at 50%; 1.86 at 58%	Assumed flow and consumption, plus PaO_2 estimates	△ Highly sensitive to assumptions
Aerobic supply threshold	SpO_2 64%	Output of the source model	△ Not a validated threshold
FiO_2 from 3 L/min nasal cannula	~50%	Fixed flow-to-fraction conversion	◌ Assumed
Neurological outcome	Injury “almost certainly” absent if heart rate ≥ 100 bpm	Heart rate not in the narrative; all inputs above	▨ Cannot be established from this source

Values as stated by the reconstruction. Status is assigned by the authors of this paper.

3.2 Model specification

The calculator was re-implemented from its displayed equations, labels, and intermediate values. The pulse-oximeter reading, corrected for an optional bias b, is taken as arterial saturation:

$$S_a = \frac{\mathrm{SpO_2} - b}{100} \tag{1}$$

The standard P_{50} (the PO_2 at which haemoglobin is half saturated) is shifted for arterial and venous pH. A Bohr coefficient of 0.48 reproduces the calculator’s displayed values (19 mmHg at pH 7.40 becomes 21.7 mmHg at pH 7.28 and 23.2 mmHg at 7.22):

$$P_{50,a} = P_{50} \cdot 10^{\,0.48\,(7.4 - \mathrm{pH}_a)}$$
$$P_{50,v} = P_{50} \cdot 10^{\,0.48\,(7.4 - \mathrm{pH}_a + \Delta\mathrm{pH})} \tag{2}$$

Arterial PO_2 follows by inverting the Hill equation with exponent n:

$$\mathrm{PaO_2} = P_{50,a} \left(\frac{S_a}{1 - S_a} \right)^{1/n} \tag{3}$$

A critical venous PO_2, $P_{v,\mathrm{crit}}$, labelled by the calculator as a “cerebral/coronary limit”, defines the saturation below which oxygen is treated as unavailable to tissue:

$$S_{v,\mathrm{crit}} = \frac{P_{v,\mathrm{crit}}^{\,n}}{P_{v,\mathrm{crit}}^{\,n} + P_{50,v}^{\,n}} \tag{4}$$

Oxygen contents (mL O_2/dL) use the conventional binding and solubility constants:

$$\mathrm{CaO_2} = 1.34\,\mathrm{Hb}\,S_a + 0.003\,\mathrm{PaO_2}$$
$$C_{v,\mathrm{crit}} = 1.34\,\mathrm{Hb}\,S_{v,\mathrm{crit}} + 0.003\,P_{v,\mathrm{crit}} \tag{5}$$

The extraction margin is the oxygen deliverable above the floor, per unit demand, where Q is cardiac output, f the fraction of flow treated as non-shunted, and $\dot{V}_{\mathrm{O_2}}$ whole-body oxygen demand:

$$M = \frac{Q\,f\,\left(\mathrm{CaO_2} - C_{v,\mathrm{crit}}\right)}{\dot{V}_{\mathrm{O_2}}} \tag{6}$$

Values of M below 1 are interpreted by the calculator as a hypoxic deficit. Default inputs are listed in Table 2.

3.3 Reproduction

We compared re-implemented outputs with every value displayed by the calculator at default inputs, and with each preset scenario whose inputs could be inferred. For the arterial PO_2 values quoted in the reconstruction, we also computed the Severinghaus equation[6] with PO_2 scaled to a P_{50} of 21.2 mmHg, as a cross-check on the dissociation curve that may have been used.

3.4 Local and one-at-a-time sensitivity

Local sensitivity was measured by the elasticity of M with respect to each input, by central differences at default values:

$$\varepsilon_\theta = \frac{\partial \ln M}{\partial \ln \theta}, \qquad \varepsilon_{S_a} \approx \frac{S_a}{S_a - S_{v,\mathrm{crit}}} \tag{7}$$

An elasticity of 4 means that a 1% relative change in the input changes M by about 4%. The approximation in Equation 7 neglects dissolved oxygen and follows directly from Equations 5 and 6. Each input was then varied alone across the range in Table 2, with all others at default.

3.5 Joint uncertainty

Inputs were drawn independently from normal distributions centred on the calculator's defaults, with the standard deviations in Table 2, and propagated through Equations 1-6 (20,000 draws per reading, seed 1). The oximeter standard deviation of 5.5 points is the value stated by the calculator itself for a 50% reading. The other spreads are illustrative: they are not derived from population data, and they are chosen so that a reader can judge whether the conclusion depends on them. Inputs were not correlated, which probably overstates some combinations and understates others.

Table 2. Model inputs: calculator defaults and illustrative uncertainty

INPUT	DEFAULT	SD (MONTE CARLO)	ONE-AT-A-TIME RANGE	BASIS IN THIS CASE
Oximeter bias	0 points	5.5	-5.5 to +5.5	SD as stated by the calculator itself at SpO_2 50%
Standard P_{50}	19 mmHg	0.75	18 to 20	Assumed
Hill exponent	2.7	0.1	2.5 to 2.9	Assumed
Arterial pH	7.28	0.05	7.20 to 7.36	Not measured in the interval
A-V pH difference	0.06	0.02	0.03 to 0.09	Assumed
Critical venous PO_2	19.5 mmHg	1.5	17 to 22	Assumed
Haemoglobin	16 g/dL	1.2	14 to 18	Not measured in the interval
O_2 demand	8.5 mL/kg/min	1.2	7 to 10	Assumed
Cardiac output	160 mL/kg/min	25	130 to 190	Assumed
Non-shunted flow fraction	0.9	0.05	0.8 to 1.0	Assumed; truncated to 0.6-1.0

Cumulative oxygen debt was computed as in the reconstruction:

$$D = \sum_i \dot{V}_{O_2}\ \max(0,\ 1 - M_i)\ \Delta t_i \tag{8}$$

with phases at 50% (start of the assumed deficit to minute 17), 58% (minutes 17-27), and a 1.5-minute transition at 61%. We varied the assumed start of the deficit from minute 5 to minute 17.

3.6 Literature verification

Statements in the reconstruction that rely on external physiology or guidance were checked against published sources where possible. Network restrictions in the drafting environment prevented full-text retrieval. References 1-9 were therefore verified against bibliographic records and abstracts, and statements attributed to them are limited to what those records support.

4 Results

4.1 The model was reproduced, except at 94%

The re-implementation matched every displayed default value (Table 3). At a 50% reading the arterial content is 10.79 mL/dL and the critical venous content 8.32 mL/dL, so only 2.47 mL/dL counts as extractable. That is 23% of the arterial content, although total delivery is 1.83 times demand (Figure 1). The margin is 0 below a reading of 38.5%, reaches 1.0 at 65.9%, and reaches 1.5 at 79.6%. These values correspond to the “~64%” threshold in the reconstruction and to the calculator’s own description of its curve.

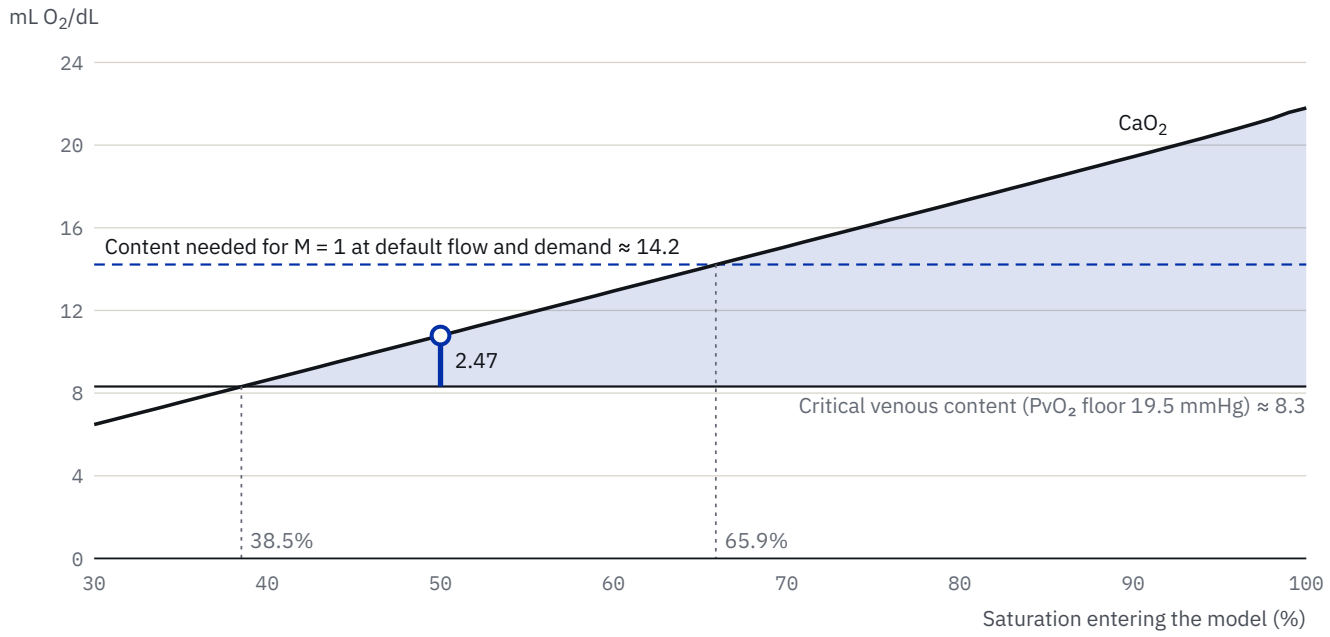
The preset for a 94% reading displays M = 2.56, but the default inputs give 2.04. Reproducing 2.56 requires an effective flow of about 181 mL/kg/min rather than 144, or an equivalent change elsewhere. The inputs behind the preset are not displayed. The arterial PO_2 values quoted in the reconstruction (21.2, 24.4 and 57.5 mmHg) also differ slightly from the calculator’s (21.7, 24.4 and 60.1 mmHg), which suggests that versions or parameters changed between the two documents.

Table 3. Reproduction of displayed calculator values

QUANTITY	READING	CALCULATOR / RECONSTRUCTION	REPRODUCED	STATUS
Arterial P_{50} / venous P_{50} (mmHg)	50%	21.7 / 23.2	21.7 / 23.2	Exact
Critical venous saturation	50%	38.5%	38.5%	Exact
CaO_2 / critical venous content / extractable (mL/dL)	50%	10.79 / 8.32 / 2.47	10.79 / 8.32 / 2.47	Exact
Total delivery / demand	50%	1.83	1.83	Exact
Extraction margin M	50%	0.42	0.418	Exact
Extraction margin M	58%	0.71	0.709	Exact
Extraction margin M	94%	2.56	2.04	Not reproduced
Reading at which M = 0 / 1 / 1.5	n/a	~38% / ~66% / ~80%	38.5% / 65.9% / 79.6%	Exact
PaO_2 (mmHg): calculator	50 / 58 / 94%	21.7 (50% only)	21.7 / 24.4 / 60.1	Hill, n = 2.7
PaO_2 (mmHg): reconstruction	50 / 58 / 94%	21.2 / 24.4 / 57.5	21.2 / 24.0 / 56.1	Severinghaus, approximate

Reproduced values use Equations 1-6 at the default inputs in Table 2. The Severinghaus column scales PO2 to a P50 of 21.2 mmHg.

Figure 1. **Where the margin comes from**



Arterial oxygen content (solid) against the saturation entering the model, at default inputs. Only the shaded region above the critical venous content counts as extractable. At a 50% reading, that gap is 2.47 mL/dL. The margin reaches 1 where arterial content reaches the dashed line, near 65.9%. Below 38.5%, nothing is counted as extractable.

4.2 The margin amplifies error near its threshold

Equation 7 explains the model's behaviour. Near the floor, the extractable content is a small difference between two large, similar quantities, so relative errors in either one are magnified. At a 50% reading, with a critical venous saturation of 38.5%, the approximate elasticity with respect to saturation is 0.50 / (0.50 − 0.385) ≈ 4.3, and the numerical value is 4.4 (Table 4). Inputs that move the floor are amplified similarly: standard P_{50} (+5.6) and critical venous PO_2 (−5.6). Arterial pH acts through both curves. Because a lower pH raises the venous P_{50} and lowers the critical venous saturation, a 0.1 fall in pH *increases M* by about 0.26 at this reading. By contrast, haemoglobin, cardiac output, the non-shunted fraction, and demand enter linearly, with elasticities of exactly ±1.

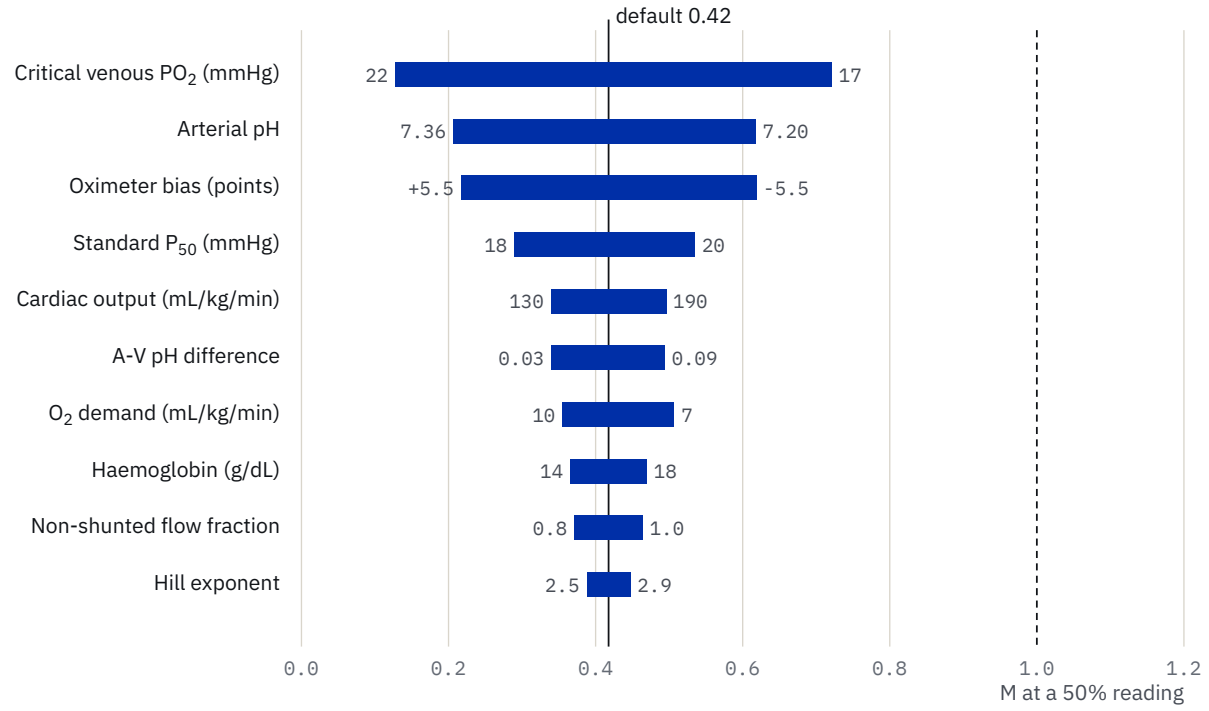
Table 4. Local elasticity of M at a 50% reading (default inputs)

INPUT	ELASTICITY	INTERPRETATION
Standard P_{50}	+5.6	Moves both arterial PO_2 and the floor
Critical venous PO_2	−5.6	Defines the floor directly
Saturation (reading)	+4.4	Difference of near-equal contents
Hill exponent	+1.0	Shape of curve near P_{50}
Haemoglobin, cardiac output, flow fraction	+1.0	Linear scaling
O_2 demand	−1.0	Linear scaling
A-V pH difference	+0.4	Shifts the venous curve
Arterial pH	−2.6 per pH unit	Absolute sensitivity; elasticity not meaningful for pH

4.3 Single inputs can move the conclusion across the threshold

Varied one at a time across the ranges in Table 2, single inputs moved *M* at a 50% reading between 0.13 and 0.72 (Figure 2). The largest effects came from the critical venous PO_2(17-22 mmHg: 0.72 to 0.13), oximeter bias (±5.5 points: 0.62 to 0.22), and arterial pH (7.20-7.36: 0.62 to 0.21). None of these was measured in the interval concerned. At a 58% reading, a critical venous PO_2 of 17 mmHg alone raises *M* to 1.01, which removes the “deficit” at that reading.

Figure 2. **One input at a time, at a 50% reading**



Each bar spans the margin produced as one input moves across its range in Table 2, with all others at default. Numbers at the bar ends are the input values producing each end. The solid vertical line marks the default margin of 0.42; the dashed line marks $M = 1$.

4.4 Joint uncertainty

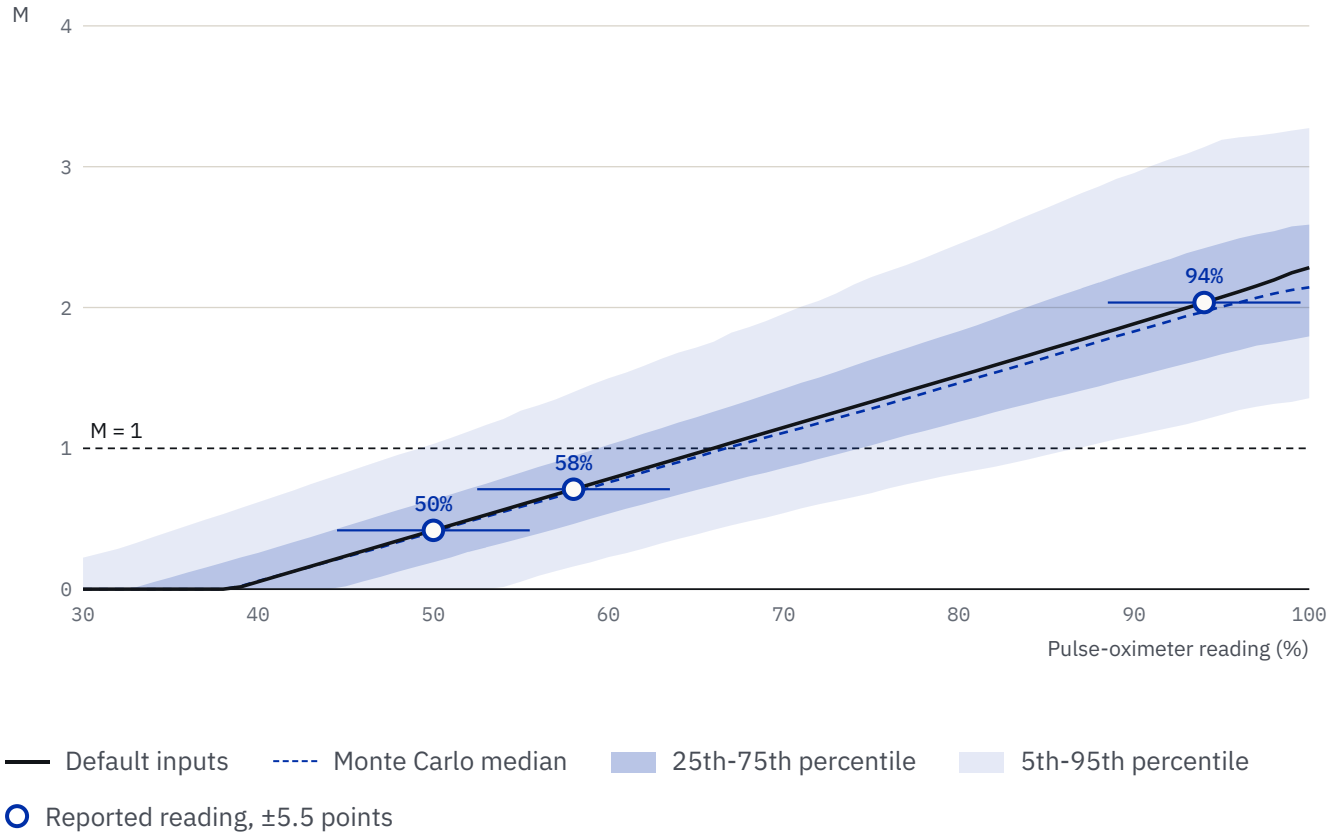
Under joint uncertainty (Table 5, Figure 3), the qualitative conclusion at the lowest reading is fairly stable: 94% of draws give $M < 1$ at 50%. At 58% the proportion falls to 79%, and the 90% interval spans 1. At 94%, 99% of draws give $M > 1$, but the value is uncertain to within roughly a factor of 2.5. The precise margins quoted in the reconstruction, such as 0.37-0.42, 0.71, and 2.56, sit inside much wider intervals than their two-decimal presentation implies.

Table 5. **Monte Carlo propagation of input uncertainty (20,000 draws per reading)**

REPORTED READING	CALCULATOR M	MEDIAN M	90% INTERVAL	SHARE WITH M < 1	SHARE WITH M < 1.5
50%	0.42	0.40	0.00-1.03	94%	>99%
58%	0.71	0.69	0.18-1.38	79%	97%
94%	2.56	1.99	1.24-3.11	1%	16%

Input distributions in Table 2. The spreads are illustrative; the proportions should be read as statements about the model under those inputs, not as probabilities about the infant.

Figure 3. **The margin across readings, with propagated uncertainty**



Margin *M* against the pulse-oximeter reading. Shaded bands are Monte Carlo percentiles (1,500 draws, computed in the browser, seed 7; Table 5 uses 20,000 draws). The three reported readings are drawn as rings with horizontal bars of ±5.5 points, the calculator's own stated oximeter uncertainty. Note how much of the band lies on both sides of *M* = 1 between readings of about 50% and 75%.

4.5 Quantities determined by assumptions

Several headline quantities in the reconstruction are fixed by assumptions rather than by observations.

The exposure window. The 22-24-minute exposure begins at minute 5, but no saturation is reported before minute 17. In a reference cohort of 468 infants who received no delivery-room intervention, the median time to reach SpO_2 above 90% was 7.9 minutes (interquartile range 5.0-10)[1], so values below 90% at minute 5 are common in uncomplicated transition. The start of the window is therefore an assumption. Current guidelines give target ranges for infants receiving resuscitation that rise from 80-85% at 5 minutes to 85-95% at 10 minutes[2]. They describe a target, not the trajectory of any individual infant.

The cumulative debt. With default inputs, Equation 8 gives about 86 mL O_2/kg if the 50% plateau starts at minute 5. This is close to the reconstruction's 91. The debt falls to 62 if the plateau starts at minute 10, 37 at minute 15, and 27 if the deficit is counted only from the first reading. Under joint uncertainty, the 90% interval was 0-192 mL O_2/kg for a minute-5 start, and 0-84 for a minute-17 start. The concept also assumes that unmet demand accumulates as debt instead of being met by reduced consumption. The record contains no lactate or blood gas value from the interval to test this.

The inspired oxygen fraction. The reconstruction equates 3 L/min by nasal cannula with an FiO_2 of about 50%. Published methods estimate effective FiO_2 from cannula flow, oxygen concentration, and the infant's breathing (inspiratory time, tidal volume, and the fraction of nasal breathing)[4]. A bench study found that commonly used formulas misestimate the delivered fraction as ventilation changes[5]. Without the infant's weight and breathing pattern, the effective FiO_2 is unknown. It does not, in any case, enter the margin calculation at 94%, which uses the reading directly.

The cerebral margin. The reconstruction's cerebral margins have the same structure as Equation 6. Its values of 1.02 at 50% and 1.86 at 58%, with cerebral blood flow of 85 mL/100 g/min and cerebral oxygen consumption of 1.8 mL/100 g/min, are reproduced by a haemoglobin of about 16.6 g/dL and a floor saturation of about 0.40. The margin is proportional to the assumed blood flow: without the assumed doubling it is 0.48. An assumed cerebral blood flow therefore separates "adequate" from "inadequate". Cerebral blood flow requirements in newborns differ from those in adults[9], so adult-derived values should not be assumed to transfer.

4.6 The record itself is uncertain

The two summaries of the same record give different first readings (Section 2.1). If the first documented value was 58%, the 50% plateau from minute 5, and the largest component of the cumulative debt, rest on a reading that may not exist. If both readings exist, their order and timing within minutes 17-27 are not established. Either way, the record determines the result more than any model input. Resolving the discrepancy requires the primary record, and no amount of modelling can replace it.

4.7 The earlier calculator

As described by the reconstruction, the earlier calculator converted a 50% pulse-oximeter reading into a 71.1% saturation "because of fetal haemoglobin" and assumed a cardiac output of 400 mL/kg/min, giving "5.67x SAFE". The conversion mixes two quantities. Fetal haemoglobin shifts the relationship between PO_2 and saturation, but a pulse oximeter reports saturation. A systematic review of ten neonatal studies found SpO_2-SaO_2 bias between about −3.6 and +4.2 percentage points, and found that higher fetal haemoglobin altered the PO_2-SpO_2relationship[3]. A correction of 21 percentage points lies far outside that range. In our reproduction, applying an adult curve to obtain PO_2 and then a fetal curve to obtain saturation gives 69-75% for fetal P_{50} values of 18-20 mmHg. This is the likely origin of the 71.1%, and it counts the same left shift twice. The 400 mL/kg/min output is 2.5 times the v2 default of 160. The v2 calculator corrected both. Our results show that its own outputs still need intervals.

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5 Discussion

5.1 Principal findings

The calculator is a coherent, inspectable model, and we could reproduce it from its own display. That matters: most AI-generated clinical reasoning cannot be reproduced at all. Its qualitative signal at the lowest reading, that oxygen above a critical venous floor was less than assumed demand, held in most of the input space we

explored. Its quantitative outputs did not. The margins are reported to two decimal places, but they belong to wide intervals that, at 58%, straddle the threshold that gives them meaning. The cumulative debt and the length of the exposure are set mainly by an assumption about an interval with no recorded saturation. On the record supplied, it is also uncertain which readings exist.

5.2 Why near-threshold margins mislead

The amplification in Section 4.2 is a general property, not a defect of this calculator. Any index defined as a difference above a floor, then divided by a demand, will be most sensitive exactly where clinicians most want to use it: near the floor. Reporting such an index without propagated uncertainty invites false precision. The calculator's own Monte Carlo panel acknowledges this. Our contribution is to show how large the effect is at the readings in question.

5.3 Isolated hypoxaemia and hypoxia-ischaemia

The reconstruction distinguished isolated hypoxaemia, low arterial oxygen with preserved circulation, from hypoxia-ischaemia, in which blood flow also fails. The distinction is established in experimental work: the classic neonatal rodent model produces injury by combining carotid ligation with hypoxia, and uses the hemisphere exposed to hypoxia alone as an internal comparison[7,8]. That framework makes circulatory status central. In this case, circulatory status is not documented: heart rate is not reported for the interval, and the reconstruction's inference of intact circulation from the later response to oxygen is plausible but indirect. Evidence from rodents also does not set thresholds for human infants. A conclusion about neurological outcome cannot be drawn from this record or this model. That question belongs to clinical assessment.

5.4 What we did not assess

We did not assess whether care met any standard. Doing so would require the primary record, the clinical context, and expert clinical review, none of which a model can supply. Guideline target ranges[2] are cited above only to show what such targets describe.

5.5 Implications for computational clinical reasoning

The case shows both sides of the opportunity. An AI-assisted analysis turned four or five reported details into an explicit, mechanistic, and reproducible model, which is more than most retrospective reasoning achieves. The same fluency turned assumptions into apparent findings. We propose eight practices for any computational reconstruction of a clinical event:

1. **Separate evidence classes.** Label every value as measured, reported, inferred, assumed, or unknown.
2. **Preserve gaps.** Never interpolate through an unrecorded interval without labelling the interpolation as an assumption.

3. **Publish the model.** State equations, parameters, and software versions so that results can be reproduced.

4. **Report intervals, not points.** Propagate measurement error, including device error at low saturation.

5. **Show sensitivity near thresholds.** Identify the inputs that can move a conclusion across a decision boundary.

6. **Reconcile the record first.** Resolve conflicting observations before modelling them.

7. **Name the decisive measurement.** State which observation would most change the conclusion.

8. **Keep verdicts out of models.** Outcome, causation, and standard of care require clinical and expert judgement.

6 Limitations

- The primary clinical record was not reviewed; all case details are reported, and the record contains at least one unresolved discrepancy.
- The calculator was re-implemented from its displayed equations and values, not from source code; the Bohr coefficient was inferred from displayed values, and the inputs behind the 94% preset were not available.
- Input spreads for the Monte Carlo analysis are illustrative and independent; real inputs are correlated, and population-derived distributions could narrow or widen the intervals.
- The model is a single-compartment abstraction; regional flow, shunt, and heterogeneous critical thresholds are not represented, and the model itself has not been validated against outcomes.
- Literature was verified at the level of bibliographic records and abstracts because of network restrictions; full-text review is planned for a subsequent version.
- The analysis was drafted with AI assistance (see Declarations); all calculations were independently reproduced by script.

7 Conclusion

An AI-assisted physiological reconstruction of a neonatal oxygenation timeline could be reproduced, interrogated, and stress-tested. Its qualitative signal at the lowest reading was fairly robust. Its precise margins, threshold, cumulative debt, and neurological conclusion were not supported by the record. The value of computational reasoning in such cases lies less in the verdicts it produces than in making its dependence on assumptions explicit, and in identifying the measurements that would settle the question.

Declarations

Ethics and privacy

Case materials were supplied privately for research use. Names, places, dates, clock times, and record page references have been removed, and times are expressed in minutes after birth.

Clinical use

Nothing in this paper is clinical guidance. The model and calculator discussed are not clinical decision tools and have not been validated.

Use of AI

The source reconstruction was generated with the assistance of a large language model. This paper was drafted with AI assistance under the direction of the authors; all quantitative results were produced by the reproduction script, and all references were checked against bibliographic records.

Code and data

Equations 1-8 fully specify the model. A dependency-free Python reproduction script regenerates Tables 3-5 and the debt analysis. Figures are computed in the browser from the same equations.

How to cite

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